Spectral estimates for Conway tangle sources

Abstract

We bound the exact colored source of a fixed strict Conway composition, uniformly over its physical circle colors, retained half twists and all sufficiently large integer levels. The upper exponential rate is the maximum of its positive complete-color action. A color a fixed fraction below complete color gives a strict loss, as does the entire difference between the full source and its selected core. The proof retains the full root algebra, including first derivatives at repeated spectral roots. An exact Fourier formula controls a character before a turn; a positive comparison with shared indices controls regions without a character. Two supporting-plane inequalities, proved by outward rational arithmetic, propagate this comparison. We also prove the uniform signed recurrence estimates for the local core coefficients. The result includes strict compositions with an elementary closing leaf and the isolated three-leaf closure. A separate corollary treats the strict two-leaf closure. These are source bounds before filling, not a filled invariant or volume asymptotic.

1 The source and the result

A circle surrounding two strands gives a polynomial operator before it is filled by twists. Stacking tangles multiplies these operators, a quarter turn acts by a specified linear map, and cylindrical closure takes a specified linear functional. The spectral polynomial has repeated roots: values alone do not determine the operator. Its first derivatives are needed both in the original finite algebra and in the cancellation that bounds a small physical color.

Separate absolute bounds at every intermediate operation are inadequate. A turn can have an exponential norm cost which need not be the intrinsic rate of that subtree. We instead alternate two estimates. A character followed by a turn admits a Fourier estimate with its exact endpoint currents. A region without a character admits a positive comparison whose shared indices determine the correct cost. The variational identities for those two operations retain their additive constants.

1.1 Constructions and their complete action

The diagrams use ordered products of two-strand boxes, their clockwise quarter turns and cylindrical closure. Each elementary box is $h_i = \mathcal{T}(R^{\epsilon_i} L_{j_i})$, where L_{j_i} is its circle operator, $1 \le j_i \le N$, and $\epsilon_i \in \{0, 1, -1\}$ specifies no retained crossing or one crossing of the indicated sign. Every box occurs once. The construction and its number $H \ge 3$ of boxes are fixed before the level N tends to infinity. A physical color is an integer dimension; it is not a continuous Fourier parameter.

Strictness is a finite condition on this syntax. The types L, H, A denote output intervals $(0, 1/2), (1/2, 1), (0, 1)$, respectively. An elementary box has type L. A turn interchanges L, H and preserves A. With either input order the product table is

$$LL, LA, AA \longmapsto A, \qquad LH, HA \longmapsto H, \qquad HH \longmapsto \varnothing.$$

At closure the allowed pairs are AA, AL, AH, LH and their reversals. A presentation passing this test is called *strict*. The empty type means absence of a strict real state, not necessarily a zero finite operator. The type letter H is unrelated to the box count.

Write $Z_N(\mathbf{j}, \boldsymbol{\epsilon})$ for the original scalar, with the single reduced cut and normalization of [1]. Section 2 specifies the finite operators and each exact core–alias split. Let Z_N^{core} be the same contraction with every box replaced by its core. Thus these are also the full and core sources formerly denoted by $\mathcal{A}_N, \mathcal{C}_N$. Let P be the limiting pure state polytope, retaining every shared wire index and the turn and closing relations. The closing product has coefficient one. With the explicit continuous functions g, m in (2.12)–(2.13), its complete action is

$$F_0 = \max_{y \in P} \left\{ \sum_{i=1}^{H} g(a_i(y)) + \sum_{v=1}^{H-2} m(x_v(y), y_v(y), z_v(y)) \right\}. \tag{1.1}$$

The strict profile theorem [3] gives a unique interior maximum on the primitive state space. Its value is also the rate of $Z_N(\mathbf{N}, \mathbf{0})$, which is positive for every sufficiently large integer N.

Theorem 1.1 (Whole-source and color bounds)**.** *Fix a strict presentation with $H \geq 3$. Uniformly over all its physical colors and retained signs,*

$$|Z_N(\mathbf{j}, \boldsymbol{\epsilon})| \leq \exp\{N(F_0 + o(1))\}. \tag{1.2}$$

For every fixed $\delta > 0$ there are $\gamma_\delta > 0, N_\delta$ such that

$$\max_i (N - j_i) \geq \delta N \quad \Longrightarrow \quad |Z_N(\mathbf{j}, \boldsymbol{\epsilon})| \leq \exp\{N(F_0 - \gamma_\delta)\}, \qquad N \geq N_\delta. \tag{1.3}$$

There are also $\gamma > 0, N_0$ such that

$$|Z_N - Z_N^{\mathrm{core}}| \leq \exp\{N(F_0 - \gamma)\}, \qquad N \geq N_0. \tag{1.4}$$

All sufficiently large integer levels are included, with no parity or primality restriction. Constants depend on the fixed presentation; no bound for constructions growing with N is asserted.

This includes the earlier completed-root grammar: start with a product of two elementary boxes, allow turns, products of completed boxes and single-box attachments, and close two completed boxes. The strict class also allows an elementary closing leaf. Corollary 9.1 handles the strict two-leaf closure separately, with rate $2G_0$, where $G_0 = g(1/4)$. The final alias gap follows from the full physical-color bound, not by identifying high Newton indices with complementary colors. A crossing can mix both Newton ranges.

1.2 The proof and its inputs

The exact finite quotient, turn, pairing, product and circle formulas are [1, Proposition 3.2, Theorem 3.3 and Proposition 4.1]. The crossing values and first derivatives are also stated in [2, Proposition 5.1]. We prove the needed complete-node conditioning directly in Section 2.3, including all high-index normals. No moving-meridian or circle-tree estimate is an input.

Section 3 retains the local signed recurrence estimate for applications that require individual core coefficients. It is independent of the whole-source proof below, and asserts no polynomial error at colliding roots. Sections 4–5 prove two scalar supporting-plane inequalities; Section 10 supplies their full rational certificates. Their parameter hypotheses are stated explicitly. The comparison for arbitrary arrays in Section 8 uses those scalar inequalities, not a completed-root source theorem applied outside its grammar.

The finite strict-state, reduced-junction and pure boundary-profile results of [3] identify the component decomposition, stationary indices and absolute rate constants. The completed-root boundary inequalities in [1, Lemma 6.1] are used only for the completed pure references specified below. Neither companion's quantum core asymptotic is an input. The Fourier and component proofs in Sections 6–9 then establish the whole-source theorem, including the three-leaf reference at the endpoint of the two-leaf comparison.

A later filling theorem still needs a normalized nonzero core asymptotic, the exact oscillatory filling transformation and a marked geometric identification of the selected action. None follows from an upper source bound alone.

2 Exact finite coefficients and their supports

Put $d = N - 1$, $M = \lfloor d/2 \rfloor$, $q = e^{\pi \mathrm{i}/N}$ and $Q = q^2$. Throughout this paper

$$S_r = \prod_{a=1}^{r} 2\sin(\pi a/N), \qquad S_0 = 1, \qquad c_a = q^{2a+1} + q^{-2a-1}.$$

Thus S_r is a product, not a single sine. The root algebra is

$$\mathscr{A}_N = \mathbb{C}[z]/(P_N(z)), \qquad P_N(z) = 2T_N(z/2) + 2 = \prod_{a=0}^{d} (z - c_a),$$

where $T_N(\cos t) = \cos Nt$. Repeated entries in the displayed product remain repeated roots. Equality in $\mathscr{A}_N$ preserves values and first derivatives at its double roots; at odd N the simple root -2 also remains. The normalized Newton basis is

$$e_r(z) = S_r^{-2} \prod_{a=0}^{r-1} (c_a - z), \qquad 0 \le r \le d. \tag{2.1}$$

The physical clockwise turn $\mathcal{T}$ and normalized cylindrical closure ϕ satisfy

$$\mathcal{T} e_r = e_{d-r}, \qquad \phi(e_i e_j) = \mathbf{1}_{i+j=d}, \qquad \phi(F) = (-1)^d N^2 [z^d](F \bmod P_N). \tag{2.2}$$

Coefficient reversal makes the turn adjoint for this pairing:

$$\phi((\mathcal{T}F)G) = \phi(F\mathcal{T}G), \qquad \phi((\mathcal{T}F)(\mathcal{T}G)) = \phi(FG). \tag{2.3}$$

It is a linear involution, not an algebra homomorphism. A stack is multiplication. For $i + j \le d$ its exact coefficients are

$$e_i e_j = \sum_{k=\max(i,j)}^{i+j} \frac{S_{i+j} S_k}{S_i S_j S_{k-i} S_{k-j} S_{i+j-k}} e_k. \tag{2.4}$$

For $i + j \ge N$ the product is zero in the full algebra, with multiplicities. All nonzero coefficients in (2.4) are positive. We call one use of this multiplication a *gate*. Its exact finite support is

$$i, j \ge 0, \qquad i + j \le d, \qquad \max(i, j) \le k \le i + j. \tag{2.5}$$

At a closing gate the trace forces $i + j = d$, and its coefficient is one. This is an exact constraint before any scaling by N.

2.1 The core and the alias

Let $U_0(z/2) = 1$, $U_1(z/2) = z$, and $U_{r+1}(z/2) = zU_r(z/2) - U_{r-1}(z/2)$. The ordinary circle is $L_j = (-1)^{j-1}U_{j-1}(z/2)$, and the elementary turned piece is $h_j = \mathcal{T}L_j$. Put $h_0 = 0$. For a high color $j = N - h$, $0 \le h \le M$, its exact split is

$$h_{N-h} = C_{N-h} + A_{N-h}, \tag{2.6}$$

$$C_{N-h} = \sum_{v=h}^{M} \frac{N^2 S_{2v}}{S_v^2 S_{v-h} S_{v+h}} e_v, \qquad A_{N-h} = (-1)^N h_h. \tag{2.7}$$

For the remaining physical colors $1 \le j \le \lfloor N/2 \rfloor$, set $C_j = 0$ and $A_j = h_j$. The low-color formula is

$$h_j = \sum_{r=0}^{j-1} (-1)^{j-1+r} \frac{S_r^2 S_{j+r}}{S_{2r+1} S_{j-r-1}} e_{d-r}. \tag{2.8}$$

There is no overlap or omitted color in this definition. The high and low *colors* should be distinguished from the high and low *Newton indices*: the high-color core has indices at most M, while an alias before the retained half twist in (2.8) has indices $d - r \ge N/2$.

Let $\kappa = q^{d^2/2}$, with $q^a = e^{\pi \mathrm{i} a/N}$ for real a. A retained half twist acts by

$$\Theta_\epsilon e_v = \kappa^\epsilon (-1)^v \sum_{r=0}^{v} q^{\epsilon v(r+1)} \frac{S_v}{S_r S_{v-r}} e_r, \qquad \epsilon = \pm 1, \qquad \Theta_0 = 1. \tag{2.9}$$

Define $C_j^\epsilon = \Theta_\epsilon C_j$ and $A_j^\epsilon = \Theta_\epsilon A_j$. The core still has indices at most M, but a half-twisted alias can occupy every index from 0 to d. Thus the alias must be estimated on the entire gate support (2.5). The phases in (2.9) are retained until after the finite sum has been formed.

The transformation is exactly $\Theta_\epsilon F = \mathcal{T}(R^\epsilon \mathcal{T} F)$. The physical crossing has

$$R(c_k) = \kappa (-1)^k q^{k(k+1)}, \qquad \frac{R'(c_k)}{R(c_k)} = -\frac{d - 2k}{4\mathrm{i} \sin((2k+1)\pi/N)}. \tag{2.10}$$

The derivative is required only at double nodes; at the odd simple node its value suffices. The inverse crossing is the inverse of this class in the full quotient.

Equations (2.2)–(2.9) are the physical identities imported above. Their derivation in [1] is in the full quotient, not merely at the distinct roots. Consequently the estimates below concern the original normal data too; no semisimple replacement of a circle is made.

2.2 Continuous actions and elementary bounds

Define the continuous function

$$J(t) = \int_0^t \log(2 \sin \pi u)\, \mathrm{d}u, \qquad 0 \le t \le 1. \tag{2.11}$$

It satisfies $J(1 - t) = -J(t)$ and $J(0) = J(1/2) = J(1) = 0$. The leaf and gate actions are

$$g(x) = J(2x) - 4J(x), \qquad 0 \le x \le \tfrac{1}{2}, \tag{2.12}$$

$$\begin{aligned} m(x, y, z) = {} & J(x + y) + J(z) - J(x) - J(y) \\ & - J(z - x) - J(z - y) - J(x + y - z). \end{aligned} \tag{2.13}$$

The full limiting gate polytope is

$$D = \{x, y \ge 0 : \ \max(x, y) \le z \le x + y \le 1\}. \tag{2.14}$$

It includes inputs larger than $1/2$. The two majorant profiles will be

$$\widetilde{g}(x) = g(\max(x, 1/4)) \quad (0 \le x \le 1/2), \qquad k(x) = -2J(x/2) \quad (0 \le x \le 1). \tag{2.15}$$

For reference, the uniform sine-factorial estimate is

$$\log S_r = NJ(r/N) + O(\log N), \qquad 0 \le r < N. \tag{2.16}$$

To see endpoint uniformity, subtract $\log t + \log(1-t)$ from the integrand of J. The remainder extends with a bounded derivative to $[0,1]$, so its Riemann sum has bounded error. The two removed sums are logarithms of ordinary factorials; integral comparison, or Stirling's inequalities, bounds their errors by $O(\log N)$ uniformly in the endpoint. This includes $r = 0$. Applied to (2.4), it gives the exponent $Nm(i/N, j/N, k/N)$ with a uniform $O(\log N)$ error.

All three profiles $\widetilde{g}, k, m$ are concave on their indicated domains. Indeed

$$\begin{aligned} g'(x) &= 2\log\cot \pi x, & g''(x) &< 0, \\ k'(x) &= -\log(2\sin(\pi x/2)), & k''(x) &= -\frac{\pi}{2}\cot(\pi x/2) < 0. \end{aligned}$$

The derivative of $\widetilde{g}$ is zero on its plateau and joins g' continuously at $1/4$. For the gate put $p = z - x$, $q_1 = z - y$, $r = x + y - z$, $w = 1 - x - y$. Then

$$\pi m = V(y, x, w) + V(p, q_1, 2r + w) + V(p + q_1, r, r + w), \qquad V(a, b, c) = -\pi[J(a) + J(b) + J(c)]. \tag{2.17}$$

Each triple is nonnegative and sums to one. On such a triangle V is concave: its two-variable negative Hessian is a positive multiple of $\left(\begin{smallmatrix} \cot \pi a + \cot \pi c & \cot \pi c \\ \cot \pi c & \cot \pi b + \cot \pi c \end{smallmatrix}\right)$, whose diagonal entries are positive and whose determinant is one. Continuity handles the boundary. Identity (2.17), obtained by $J(1-t) = -J(t)$, proves concavity on the full D.

2.3 Conditioning at the complete spectral nodes

The canonical representative of a quotient class has degree less than N. Off the spectral nodes we specify whether we use this representative or an actual unreduced product. Define the norm

$$\|F\|_{\mathcal{J}} = \max\big\{|F(c)| + |F'(c)| : c \text{ a double node; } |F(-2)| \text{ if } N \text{ is odd}\big\}.$$

It is a norm on the full quotient and is submultiplicative because $|FG| + |(FG)'| \le (|F| + |F'|)(|G| + |G'|)$ at each double node. The derivative at an odd simple node is not part of the data. Put

$$w_{N,r} = \begin{cases} e_r(-2), & 0 \le r \le M, \\ 1, & M < r < N. \end{cases} \tag{2.18}$$

These weights are positive. The following estimates are purely at the complete main color; no tree or filling-power hypothesis is needed.

Lemma 2.1 (Complete-node conditioning)**.** *There are absolute constants C, A such that*

$$|\phi(F)| \le CN^A \|F\|_{\mathcal{J}}, \qquad \|R^\epsilon L_j\|_{\mathcal{J}} \le CN^A, \quad 1 \le j \le N, \quad \epsilon \in \{0, 1, -1\}, \tag{2.19}$$

$$\|e_r\|_{\mathcal{J}} \le CN^A w_{N,r}, \qquad 0 \le r < N, \tag{2.20}$$

$$\log w_{N,r} = Ng(r/N) + O(\log N), \qquad 0 \le r \le M. \tag{2.21}$$

For $r > M$, every value of e_r at a spectral node is zero, but its first normals satisfy $\|e_r\|_{\mathcal{J}} \le N^2/16$. Every canonical polynomial also satisfies

$$|F(2\cos(\theta + \mathrm{i}y))| \le CN^A e^{N|y|} \|F\|_{\mathcal{J}}, \qquad \theta, y \in \mathbb{R}. \tag{2.22}$$

This last assertion is not made for arbitrary unreduced representatives.

Proof. The polynomial with the distinct complete nodes as simple roots is

$$Q_N(z) = \begin{cases} 2T_{N/2}(z/2), & N \text{ even}, \\ (z+2)\{U_M(z/2) - U_{M-1}(z/2)\}, & N = 2M+1. \end{cases}$$

Its interval norm and first derivatives are polynomial in N. At even roots, $|Q'_N| \geq N/2$. At an odd interior root $c = 2\cos t$, differentiation gives $|Q'_N(c)| = N/(2\sin(t/2))$; at -2 it gives N. Each Lagrange cardinal is bounded on $[-2,2]$ by $\|Q'_N\|_\infty/|Q'_N(c)|$, by the mean-value formula. Thus interpolation at the distinct nodes has polynomial cost.

For $\deg F < N$, interpolate its values by I with $\deg I < \deg Q_N$. Write $F - I = Q_N H$. At each double node,

$$H(c) = \frac{F'(c) - I'(c)}{Q'_N(c)}.$$

The degree of H is less than the number of double nodes, so these values determine it. The interpolating polynomial for this second step is Q_N at even levels and $Q_N/(z+2)$ at odd levels. The latter has polynomial interval norm, and its root derivatives have magnitude $|Q'_N(c)|/(c+2) \geq N/8$. The same cardinal estimate applies. Derivative bounds do not presume good conditioning: a degree-n polynomial has Chebyshev coefficients at most twice its interval norm, and $\|T'_j(z/2)\|_\infty \leq j^2/2$. This proves polynomial equivalence of the canonical interval norm, its complete value-and-normal norm, and its Chebyshev coefficient sum.

In particular $[z^d]T_d(z/2) = 1/2$, so the exact functional in (2.2) has polynomial cost. The circle polynomial has $|L_j| \leq j$ on $[-2,2]$, and the same derivative estimate gives a polynomial bound for its first derivative. The crossing has unit values and logarithmic derivative $O(N^2)$ at every double node by (2.10). Its inverse has the same bounds. This proves (2.19). The Chebyshev coefficient bound and $|T_k(\cos(\theta + \mathrm{i}y))| \leq e^{k|y|}$ prove (2.22).

For low r, the values $e_r(c)$ at distinct nodes are either zero or nonnegative, and do not exceed $e_r(-2)$. Its degree is less than the number of distinct nodes, so the first interpolation step bounds its interval norm and derivative by $CN^A e_r(-2)$. The product at -2 is

$$e_r(-2) = \frac{\prod_{a<r} 4\cos^2((a+1/2)\pi/N)}{S_r^2}.$$

Endpoint sum–integral comparison, as in (2.16), gives $\log e_r(-2) = Ng(r/N) + O(\log N)$ uniformly in $r \leq M$. The midpoint logarithmic intervals at the upper endpoint contribute only $O(\log N)$, including both parities.

For high r, put $l = N - r \leq \lfloor N/2 \rfloor$, $\theta = \pi/N$, and $B_k(z) = \prod_{a<k}(c_a - z)$. Reflection gives $B_r B_l = (-1)^N P_N$. At a double node indexed by $0 \leq j < \lfloor N/2 \rfloor$, the derivative of e_r vanishes unless $j < l$, and otherwise

$$|e'_r(c_j)| = \frac{S_{l-1}^2}{2\sin((2j+1)\theta) S_{l+j} S_{l-1-j}}. \tag{2.23}$$

Compare quadratic coefficients in the reflected identity, using

$$\frac{|P''_N(c_j)|}{2} = \frac{N^2}{4\sin^2((2j+1)\theta)}, \qquad |B'_l(c_j)| = \frac{S_{l+j} S_{l-1-j}}{2\sin((2j+1)\theta)},$$

and $S_r S_{l-1} = N$. The ratio $D_j = S_{l+j} S_{l-1-j}/S_{l-1}^2$ starts at $2\sin(l\theta)$ and is nondecreasing, since

$$\frac{D_{j+1}}{D_j} = \frac{\sin((l+j+1)\theta)}{\sin((l-1-j)\theta)} \geq 1;$$

the two angles sum to at most π. Consequently (2.23) is at most $1/[4\sin((2j+1)\theta)\sin(l\theta)] \leq N^2/16$. The odd simple-node value is zero. This retains every normal and completes the proof. □

Throughout the estimates, a uniform error $o(1)$ may absorb powers of N, a fixed number of sums and the errors of fixed estimates. It is uniform over all indicated physical colors and retained signs. A strict exponential loss is reduced before those errors are absorbed.

3 Uniform bounds for one actual leaf

The difficult coefficient is a half-twisted core. Taking the absolute value of each term in (2.9) would lose the cancellation needed for its correct exponential rate. We derive a recurrence for that signed finite sum, including its endpoint condition, and then bound products of its recurrence matrices.

Proposition 3.1 (Leaf envelopes)**.** *There are a sequence $e_N \to 0$ and an absolute constant C such that, uniformly over all physical colors and $\epsilon \in \{0, 1, -1\}$,*

$$|[e_r]C_j^\epsilon| \le \exp\{N[\widetilde{g}(r/N) + e_N]\}, \qquad 0 \le r \le M, \tag{3.1}$$

$$|[e_r]A_j^\epsilon| \le N^C \exp\{Nk(r/N)\}, \qquad 0 \le r \le d. \tag{3.2}$$

The core is zero above M. Moreover, if $I \subset (1/4, 1/2)$ is compact and $\delta > 0$, there is $\eta > 0$ such that, for all sufficiently large N,

$$h \ge \delta N, \quad r/N \in I, \quad 0 \le h \le M \quad \Longrightarrow \quad |[e_r]C_{N-h}^\epsilon| \le \exp\{N[g(r/N) - \eta]\}. \tag{3.3}$$

Zero coefficients are included. No polynomial prefactor for the core is asserted at collisions of recurrence roots.

3.1 The finite Ward identity

Here $F_r = (Q;Q)_r = \prod_{a=1}^{r}(1 - Q^a)$, $F_0 = 1$. For integers $0 \le h, r \le M$ put

$$A_{h,r} = \sum_{v=\max(r,h)}^{M} \frac{Q^{-vr} F_{2v}}{F_v F_{v-h} F_{v+h} F_{v-r}}, \tag{3.4}$$

$$D_{0,r} = \frac{Q^{-r^2} F_{2r}}{F_r^3}, \qquad V_{h,r} = \frac{q^{h^2} A_{h,r}}{D_{0,r}}. \tag{3.5}$$

Every factorial in these sums has a nonnegative index below N, and $D_{0,r} \neq 0$. The lower endpoint is $\max(r, h)$ even when the output index is below the color deficit.

The elementary phase identity

$$F_a = (-\mathrm{i})^a q^{a(a+1)/2} S_a$$

converts the negative half-twist coefficient in (2.9) to

$$[e_r]C_{N-h}^{-1} = \frac{N^2 \kappa^{-1}}{S_r}\, \mathrm{i}^r q^{(r^2-r)/2+h^2} A_{h,r}.$$

The positive sign is its complex conjugate in the real Newton basis. Thus

$$|[e_r]C_{N-h}^\epsilon| = \frac{N^2 S_{2r}}{S_r^4} |V_{h,r}|, \qquad \epsilon = \pm 1. \tag{3.6}$$

This is the modulus of the original sum, after its phases have been combined.

Set $H = Q^h$, $Y = Q^r$ and $T_h = H + H^{-1}$. The exact finite sum satisfies

$$QY^2(1-Q^2Y)A_{h,r+2} + Y(T_hQ^3Y^2 + Q + 1)A_{h,r+1} - Q(QY+1)(QY^2-1)A_{h,r} = 0. \quad (3.7)$$

Absent sums above M are zero. Here is an explicit telescoping proof. If $z = Q^v$, the ratio of consecutive summands of (3.4) is

$$R(z) = \frac{Y^{-1}(1-Qz^2)(1+Qz)}{(1-Qz/H)(1-QHz)(1-Qz/Y)}.$$

Use

$$G_h(z) = \frac{(1-z/H)(1-Hz)(1-z/Y)}{z^2}.$$

Expanding $R(z)G_h(Qz) - G_h(z)$ gives

$$\left(\frac{1}{YQ^2} - 1\right)z^{-2} + \left(\frac{1}{YQ} + H + H^{-1} + \frac{1}{Y}\right)z^{-1} - \frac{1}{YQ} - 1 - \frac{H+H^{-1}}{Y}. \quad (3.8)$$

The ratio of a summand at $r+1$ to its value at r is $z^{-1} - Y^{-1}$. Applying this identity twice expresses the first two z^{-1} moments in terms of $A_{h,r+1}$ and $A_{h,r+2}$ and gives (3.7) after clearing the displayed denominators.

The endpoints vanish in this cleared form. At $v = \max(r, h)$ one of $1 - z/Y$ and $1 - z/H$ is zero. At the upper endpoint, multiplying the next summand by $G_h(Q^{M+1})$ first cancels the three last factorial factors; the remaining numerator contains F_{2M+2} and hence $1 - Q^N$. This order also covers $h = M$, where an uncanceled ratio would read $0/0$. No negative factorial or division by a zero endpoint is used. These observations prove the recurrence for $r < h$ as well as $r \geq h$.

Normalizing by (3.5) and writing $Z = Q^{r+1}$ gives

$$V_{h,r} = p_{h,N}(r)V_{h,r+1} + q_N(r)V_{h,r+2}, \quad (3.9)$$
$$p_{h,N}(r) = -\frac{Z^{-1}[(H+H^{-1})Z^2 + 1 + Q^{-1}]}{(1-Z)^2},$$
$$q_N(r) = -\frac{Q^{-2}Z^{-2}(1-QZ^2)(1+QZ)}{(1-Z)^2(1-QZ)}.$$

The denominators are nonzero for $0 \leq r \leq M-1$. At $r = M-1$, $q_N(r) = 0$ at both parities. We may set the unused value $V_{h,M+1} = 0$. The actual remaining terminal value is

$$|V_{h,M}| = \frac{S_M^2}{S_{M-h}S_{M+h}} = \begin{cases} 1, & N \text{ odd}, \\ \cos(\pi h/N), & N \text{ even}. \end{cases} \quad (3.10)$$

Indeed $S_aS_{N-1-a} = N$. For odd $N = 2M+1$ both products in the ratio equal N. For even $N = 2m$, $M = m-1$, the numerator is $N/2$ and the denominator is $N/(2\cos(\pi h/N))$. Thus the prescribed solution is bounded by one at the upper wall throughout the physical color range. For the rate arguments it suffices to take $N \geq 3$; the finite formulas still have their stated meaning at $N = 2$.

3.2 Both roots of the recurrence

Let $T_N(r)$ be the companion matrix of (3.9), so that $(V_{h,r}, V_{h,r+1})^{\mathsf{t}} = T_N(r)(V_{h,r+1}, V_{h,r+2})^{\mathsf{t}}$. Freeze $u = (r+1)\pi/N$ and $\chi = \pi h/N$ while $Q \to 1$. The limiting matrix $T_0(u, \chi)$ has eigenvalues $e^{-2\mathrm{i}u}v$, where

$$v^2 - bv - c = 0, \qquad c = \cot^2 u, \qquad b = (c-1)\cos^2\chi - 2\mathrm{i}\sqrt{c}\sin^2\chi. \quad (3.11)$$

We claim

$$\rho(T_0(u,\chi)) \le \rho_0(u) := \max(1, \cot^2 u), \qquad 0 < u \le \pi/2, \quad 0 \le \chi \le \pi/2. \tag{3.12}$$

For $0 < c < 1$, a root v on the unit circle would require $b = v - c/v$. Its locus in the b plane is the ellipse

$$\left(\frac{\Re b}{1-c}\right)^2 + \left(\frac{\Im b}{1+c}\right)^2 = 1.$$

At the actual b its left side is

$$\cos^4\chi + \frac{4c}{(1+c)^2}\sin^4\chi \le 1,$$

with strict inequality for $\chi > 0$. At the center the roots have modulus $\sqrt{c} < 1$. Along a segment inside the ellipse no root can cross the unit circle, proving the bound for both roots. Continuity covers its boundary. For $c > 1$ apply the same argument to v/c and $1/c$. At $c = 1$ the roots of $v^2 + 2\mathrm{i}\sin^2\chi\, v - 1$ have modulus one; at $c = 0$ they are 0 and $-\cos^2\chi$. This proves (3.12).

In particular, for $\chi_0 > 0$ and a compact interval $I \subset (\pi/4, \pi/2)$,

$$\sup_{u\in I,\, \chi_0 \le \chi \le \pi/2} \rho(T_0(u,\chi)) \le \theta < 1. \tag{3.13}$$

The endpoint $\chi = \pi/2$ is included: its repeated root has modulus $\sqrt{c} < 1$. We will bound matrix powers directly, without an eigenbasis that could become singular there.

3.3 Fixed blocks and the lower endpoint

Define

$$E(u) = \begin{cases} \dfrac{2}{\pi}\displaystyle\int_u^{\pi/4} \log\cot t\,\mathrm{d}t, & 0 \le u < \pi/4, \\ 0, & \pi/4 \le u \le \pi/2. \end{cases} \tag{3.14}$$

The recurrence implies

$$\limsup_{N\to\infty} \max_{0\le h,r\le M} \{N^{-1}\log|V_{h,r}| - E(\pi r/N)\} \le 0, \tag{3.15}$$

$$\limsup_{N\to\infty} \max_{\substack{\delta N \le h \le M \\ 0 \le r \le \beta N}} \{N^{-1}\log|V_{h,r}| - E(\pi r/N)\} \le -\gamma_{\delta,\beta} < 0, \tag{3.16}$$

for $0 < \delta, \beta < 1/2$. We use $\log 0 = -\infty$ and ignore an empty index set.

Here are the uniform product estimates. Fix a lower cutoff $a > 0$. For $a \le u_r = (r+1)\pi/N \le \pi/2$, the exact normalized matrices $B_N(r) = T_N(r)/\rho_0(u_r)$ differ by $O_a(N^{-1})$ from the bounded Lipschitz family $B(u_r,\chi) = T_0(u_r,\chi)/\rho_0(u_r)$. The factor $1 - QZ$ in (3.9) stays bounded away from zero on this interval at all sufficiently large levels.

For a bounded 2 by 2 matrix whose eigenvalues have modulus at most one, Cayley–Hamilton gives

$$\|B^L\| \le L\|B\| + L - 1 \le C_a L. \tag{3.17}$$

The coefficient of B is the polynomial $\sum_{j=0}^{L-1} \lambda_1^{L-1-j}\lambda_2^j$; the constant coefficient has modulus at most $L - 1$. The formula extends to equal roots. If both roots have modulus at most $\theta < 1$, it gives $\|B^L\| \le C_a L\theta^{L-1}$.

Choose a fixed integer L. Telescoping the product of L adjacent matrices against its frozen value gives, uniformly in χ,

$$B_N(r)\cdots B_N(r+L-1) = B(u_r,\chi)^L + O_{a,L}(N^{-1}).$$

For a prescribed $\varepsilon > 0$, choose L large enough that $C_a L < e^{\varepsilon L}$ and only then take N sufficiently large. Complete blocks cost at most $e^{2\varepsilon L}$; the incomplete blocks at the ends cost a fixed constant. Thus any normalized propagation above the cutoff costs at most $C_{a,\varepsilon} e^{2\varepsilon N}$.

For (3.16), choose $I \subset (\max(\pi\beta, \pi/4), \pi/2)$. Propagation from the upper wall to any $r \le \beta N$ crosses I. Equation (3.13) permits the same fixed L to be enlarged until every full block inside I has norm at most $e^{-\gamma_0 L}$ for some $\gamma_0 > 0$. There are $N|I|/(\pi L) + O(1)$ such blocks. Taking $\varepsilon < \gamma_0 |I|/(8\pi)$ leaves a strict negative rate after the remaining blocks have been included.

At the omitted lower endpoint, direct estimates in (3.9) give

$$\|T_N(r)\| \le C \frac{N^2}{(r+1)^2}, \qquad u_r \le a < \pi/8.$$

Indeed $|H + H^{-1}| \le 2$, $|1 - Z| \ge 4(r+1)/N$, and $|1 - QZ^2|/|1 - QZ|$ is bounded in this range. For at most $K \le aN/\pi$ consecutive indices the product is bounded by

$$\frac{C^K N^{2K}}{(K!)^2} \le \exp\{O(Na \log(1/a)) + O(\log N)\}.$$

This also covers a shorter segment beginning later, by enlarging the absolute constant so each omitted factor is at least one. Its normalized rate tends to zero with a.

Finally $\log \rho_0$ decreases to zero, so its right-endpoint Riemann sum satisfies

$$\sum_{j=r}^{M-1} \log \rho_0((j+1)\pi/N) \le NE(\pi r/N).$$

For the global assertion first fix a and ε, then L, then N, and finally make a, ε arbitrarily small. For the strict assertion fix I and its negative budget first; choose $a \log(1/a)$ and ε below that budget, then L, and finally N. Combining (3.10) with these estimates proves (3.15) and (3.16). Both branches, including their collisions, have been bounded.

The prefactor in (3.6) has rate $g(x)$ by (2.16), and

$$g(x) + E(\pi x) = g(\max(x, 1/4)) = \widetilde{g}(x). \tag{3.18}$$

This proves (3.1) for the two twisted signs. For an untwisted core, the coefficient vanishes for $r < h$ and otherwise has rate

$$g(x) - \Delta(x,t), \qquad \Delta(x,t) = J(x-t) + J(x+t) - 2J(x), \quad t = h/N,\ x = r/N.$$

For $t > 0$ and $x < 1/2$,

$$\partial_t \Delta = \log \frac{\sin \pi(x+t)}{\sin \pi(x-t)} > 0,$$

since the difference of the two sines is $2 \cos \pi x \sin \pi t > 0$. The boundary $t = x$ is integrable and follows by continuity. This proves the untwisted bound and its strict loss on a fixed high interval when $t \ge \delta$. For the twisted signs, choose β above that interval and apply (3.16); $E = 0$ there. Thus (3.3) follows at both parities.

3.4 The full alias range

It remains to prove (3.2). The absolute value of a coefficient in (2.8) is

$$b_{j,r} = \frac{S_r^2}{S_{2r+1}} \prod_{a=-r}^{r} 2 \sin((j+a)\pi/N), \qquad 0 \le r < j \le \lfloor N/2 \rfloor.$$

As a function of real j on $(r, N-r)$, the logarithm of the product is strictly concave and symmetric about $N/2$. Its maximum is at that center. Sine duplication gives

$$b_{j,r} \le B_r := 2\frac{\prod_{a=1}^{r} 2\sin(2a\pi/N)}{\prod_{a=0}^{r} 2\sin((2a+1)\pi/N)}.$$

Here $B_0 = \csc(\pi/N)$ and $B_{r+1}/B_r = \sin((2r+2)\pi/N)/\sin((2r+3)\pi/N)$. The ratio changes from at most one to at least one at most once, so a maximum occurs at an endpoint. The last value is $N/2$ for even N, and $\csc(\pi/N)$ for odd N. Since $\sin(\pi/N) \ge 2/N$,

$$b_{j,r} \le N/2. \tag{3.19}$$

A half twist contributes a coefficient ratio $S_v/(S_r S_{v-r})$, where $v \ge \max(r, N/2)$. Its rate is

$$J(w) - J(u) - J(w-u), \qquad w = v/N,\ u = r/N.$$

Differentiation in w gives $\log(\sin \pi w/\sin \pi(w-u))$; its maximum is at $w = (1+u)/2$. Reflection and duplication of J give the maximum $-2J(u/2) = k(u)$. Uniform factorial errors and at most N summands cost only a power of N. This proves (3.2) for both twists. The pure alias obeys the same bound because $k \ge 0$ and (3.19) applies. Proposition 3.1 is proved.

4 A two-leaf comparison on the whole output interval

In a completed pure reference the selected leaf range is

$$\frac{1}{4} < a < \frac{\arctan 2}{\pi} < \frac{1}{2}. \tag{4.1}$$

A pointwise comparison with a single complete leaf is insufficient throughout this range. The relevant quantity would be

$$\min_{0 \le u \le 1} \{g(a) + g'(a)(u-a) - k(u)\},$$

which need not be positive there. The gate joining two leaves supplies the missing comparison. We retain it, and its output tangent, before propagating an estimate into the rest of the tree.

At the selected pure two-leaf stack, write

$$X = \cot \pi a, \qquad \cot \pi b = X - \frac{1}{2X}, \qquad F_2(b) = 2g(a) + m(a,a,b), \qquad \lambda = 2\log(2X^2). \tag{4.2}$$

Here arccot takes values in $(0, \pi)$. The displayed expressions define the scalar parameters for every $X \in [1/2, 1]$. Completed pure references use the interior range; the three-leaf reference in Section 8 uses the endpoint $X = 1/2$.

Lemma 4.1 (Two-leaf gap)**.** *For $1/2 \le X \le 1$ and the slope in (4.2),*

$$\max_{D,\, y \le 1/2} \{k(x) + \widetilde{g}(y) + m(x,y,z) - \lambda z\} \le F_2(b) - \lambda b - \frac{1}{100}, \tag{4.3}$$

$$\max_{D} \{k(x) + k(y) + m(x,y,z) - \lambda z\} \le F_2(b) - \lambda b - \frac{1}{100}. \tag{4.4}$$

Proof. Let $P(\lambda)$ be the maximum of $g(x) + g(y) + m(x,y,z) - \lambda z$ on D with $x, y \le 1/2$. Direct differentiation at (a,a,b) gives

$$m_x = m_y = -2\log X, \qquad m_z = \lambda, \qquad g'(a) = 2\log X.$$

This point is strict in D, including the endpoints $X = 1/2, 1$. Concavity proves

$$P(\lambda) = F_2(b) - \lambda b.$$

The same maximum is obtained if either or both copies of g are replaced by $\widetilde{g}$: the selected point has $a \geq 1/4$, where their supporting tangents agree.

We give the complete finite certificate reduction. Write $H = k + \widetilde{g} + m$ for the first alias action and $H = k + k + m$ for the second. A rational feasible pure point $p = (p_1, p_1, p_3)$ gives

$$P(\lambda) \geq 2g(p_1) + m(p) - \lambda p_3.$$

For a strict rational alias point v, with $v_2 > 1/4$ in the first case, global concavity gives $H(w) \leq H(v) + \nabla H(v) \cdot (w - v)$. The full vertex sets are

$$\begin{aligned}\mathcal{V}_{\mathrm{AC}} &= \{(0,0,0), (1,0,1), (1/2,1/2,1/2), (0,1/2,1/2), (1/2,1/2,1)\},\\ \mathcal{V}_{\mathrm{AA}} &= \{(0,0,0), (1,0,1), (0,1,1), (1/2,1/2,1/2)\}.\end{aligned}$$

To verify that these lists are complete, the nonnegative coordinates $p = z - x$, $q_1 = z - y$, $r = x + y - z$, $w_0 = 1 - x - y$ obey $p + q_1 + 2r + w_0 = 1$. Thus D is a tetrahedron. Cutting by $y \leq 1/2$ removes one vertex and adds the two listed edge intersections.

Partition $\lambda/\log 2 \in [-2, 2]$ into the sixteen intervals $[-2+j/4, -2+(j+1)/4]$, $j = 0, \ldots, 15$. On each interval the rational data in [4] give one pure point and one tangent point for each alias action. For every vertex w and the two slope endpoints t, they give

$$2g(p_1) + m(p) - t\log 2\, p_3 - H(v) - \nabla H(v) \cdot (w - v) + t\log 2\, w_3 > \frac{1}{100}. \tag{4.5}$$

The expression is affine separately in the slope and in w. Its endpoint and vertex checks therefore prove the inequality on each full interval and polytope. There are $16 \cdot 2 \cdot (5+4) = 288$ inequalities. Section 10 proves the outward enclosure rules and specifies the exact data. The smallest certified lower bound is

$$\frac{21000210290230963809378576600597934432 9}{50000000000000000000000000000000000000000000} > \frac{1}{25} > \frac{1}{100}. \tag{4.6}$$

This proves both comparisons, including boundary maxima and the core plateau. □

5 A new leaf and an arbitrary completed child

A completed child is represented by its actual selected output and its output derivative. Its entire internal source remains in the calculation. Let $f_E(b)$ be its optimized pure complete action at output b, near the selected strict state. The imported strict maximum has negative Hessian in its internal variables, so the implicit function theorem gives this differentiable local profile. Its derivative is the boundary Lagrange multiplier.

The boundary variables from [1, Section 6] are

$$B = \cot \pi b, \qquad X^2 = \frac{1 + B^2}{1 + e^{-f_E'(b)}}, \qquad Y = e^{-f_E'(b)/2} X.$$

Thus $X^2 + Y^2 - B^2 = 1$ and $f_E'(b) = 2\log(X/Y)$. At a stack the two children have the same X, and their output variables satisfy

$$B_o = \frac{B_1 B_2 + X^2 - 1}{B_1 + B_2}, \qquad Y_o = \frac{Y_1 Y_2}{B_1 + B_2}.$$

For a leaf attached to a completed child these become

$$\begin{aligned} &a = \operatorname{arccot} X/\pi, \quad b = \operatorname{arccot} B/\pi, \quad c = \operatorname{arccot} B_p/\pi, \quad B_p = X - \frac{1}{X+B}, \\ &Y_p = \frac{Y}{X+B}, \qquad \mu = f'_E(b) = 2\log(X/Y), \qquad \lambda = f'_{hE}(c) = 2\log(X/Y_p). \end{aligned} \tag{5.1}$$

For a completed pure reference, both child and parent are completed. Lemma 6.1 of [1] then gives $0 < X, Y, Y_p < 1$. The child inequality $Y < 1$ gives $|B| < X$. The parent inequality $Y_p < 1$ gives

$$B > \frac{1}{2X} - X.$$

Together they force $X > 1/2$. These two distinct inequalities are needed below.

Define the pure and alias dual maxima

$$P(\mu, \lambda) = \max_{D,\, x \le 1/2} \{g(x) + m(x, y, z) + \mu y - \lambda z\}, \tag{5.2}$$

$$Q(\mu, \lambda) = \max_{D} \{k(x) + m(x, y, z) + \mu y - \lambda z\}. \tag{5.3}$$

Only the new leaf has the restriction $x \le 1/2$. The child coordinate y retains the whole interval $[0, 1]$; it is represented by its tangent, not replaced by a second leaf.

Lemma 5.1 (Single-attachment gap)**.** *Let $1/2 < X < 1$, $Y = \sqrt{1 + B^2 - X^2}$, $Y_p = Y/(X+B)$ and $0 < Y, Y_p < 1$, with the positive branch $X + B > 0$. Define a, b, c, μ, λ by (5.1). Then*

$$Q(\mu, \lambda) \le P(\mu, \lambda) - \frac{1}{2500}.$$

The pure maximum is attained at (a, b, c) from (5.1). Replacing g by $\widetilde{g}$ does not change this maximum.

Proof. At the displayed point, $m_x = -2\log X$, $m_y = -\mu$, $m_z = \lambda$ and $g'(a) = 2\log X$. Hence the pure objective is stationary and concavity proves its global maximality. Since $a > 1/4$, the same tangent applies to $\widetilde{g}$.

First suppose $1/2 < X \le 2/3$. The child gives $Y^2 = 1 + B^2 - X^2 \ge 1 - X^2 > X^2$, while $Y < 1$. Thus $-2\log 2 < 2\log X < \mu \le 0$. The parent gives separately $Y_p^2 = 1 + B_p^2 - X^2 \ge 1 - X^2 > X^2$, while $Y_p < 1$, so the same bounds hold for λ. Both slopes lie in $[-2\log 2, 0]$.

On the entire closed square, a second rational certificate proves

$$P(\mu, \lambda) - Q(\mu, \lambda) > \frac{1}{1000}. \tag{5.4}$$

Divide both slopes by $\log 2$. The resulting square is partitioned into 55 dyadic rectangles. Each rectangle carries a strict rational pure point p, with $p_1 < 1/2$, and a strict rational tangent point v for $H = k + m$. At each of its four slope corners and each of the four vertices of D, the verified inequality is

$$\begin{aligned} g(p_1) + m(p) - H(v) - \nabla H(v) \cdot (w - v) \\ + \mu(p_2 - w_2) - \lambda(p_3 - w_3) > \frac{1}{1000}. \end{aligned} \tag{5.5}$$

Concavity bounds the alias function by this tangent; the pure point bounds its maximum below. Affinity in the slopes and in w gives the full rectangle and full D. The data define each nonterminal subdivision node to have exactly four children, its four quarter-rectangles,

starting at $[-2,0]^2$. This proves coverage without gaps or overlap concerns. There are $55 \cdot 4 \cdot 4 = 880$ inequalities. Their least certified lower bound is

$$\frac{554371917600198748925179726840207633 73}{10^{40}} > \frac{1}{1000}. \tag{5.6}$$

For $X \geq 2/3$ an elementary one-leaf inequality suffices. Set

$$I(a) = \min_{0 \leq u \leq 1} \{g(a) + g'(a)(u-a) - k(u)\}.$$

For $a \geq a_* := \arctan\sqrt{2}/\pi$, the minimum is at $u = 1$ because $g'(a) \leq k'(1) = -\log 2$. Then

$$I(a) = g(a) + (1-a)g'(a) = \frac{2}{\pi}\,\mathrm{Ti}_2(X) + \log X, \qquad \mathrm{Ti}_2(X) = \int_0^X \frac{\arctan t}{t}\,\mathrm{d}t. \tag{5.7}$$

For example the two expressions have the same derivative in X, namely $X^{-1} + 2\arctan X/(\pi X)$, and their difference tends to zero as $X \downarrow 0$. This proves the constant in (5.7). The expression decreases with a.

For $1/4 \leq a \leq a_*$ the minimizing index is $u = (2/\pi)\arcsin(\tan^2(\pi a)/2)$. It exceeds a: equivalently $\tan^2 t > 2\sin(t/2)$ for $\pi/4 \leq t \leq \arctan\sqrt{2}$, which holds at $\pi/4$ and remains true because the derivative of the difference is $2\tan t \sec^2 t - \cos(t/2) > 0$ there and thereafter. Hence $I'(a) = g''(a)(u-a) < 0$. The minimum on $1/4 \leq a \leq \operatorname{arccot}(2/3)/\pi$ is consequently at its upper endpoint, where (5.7) applies.

Four alternating terms of $\mathrm{Ti}_2(2/3)$ give

$$\mathrm{Ti}_2(2/3) > \frac{1708762}{2679075}.$$

The series $2\sum_{j\geq 0}(1/5)^{2j+1}/(2j+1)$ for $\log(3/2)$, with its geometric tail, gives $\log(3/2) \leq 177391/437500$. Together with $\pi < 22/7$ these imply

$$I(a) > \frac{57269}{136687500} > \frac{1}{2500}. \tag{5.8}$$

The bound $\pi < 22/7$ follows, for instance, from the Machin enclosure used in Section 10. Thus $k(u) \leq g(a) + g'(a)(u-a) - 1/2500$ on the full alias interval. Adding the supporting plane of m and the child linear term cancels the input coefficients by stationarity. This proves the lemma in the remaining case. □

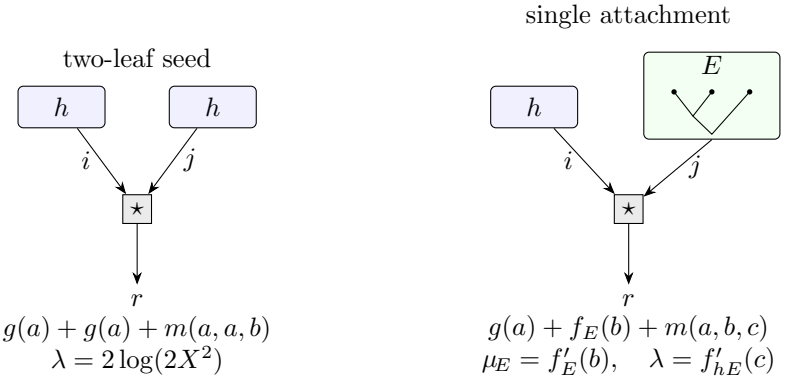


Figure 1: A completed construction retains its exact indices at every step; $\star$ is the multiplication in (2.4). In the left seed, i, j are leaf indices and r is the output of their original multiplication. In the right attachment, j is the output of the entire child E, not a new leaf. The affine estimates use the same $i/N, j/N, r/N$; the internal slopes cancel at the selected stationary state.

6 A character contraction as a Fourier functional

We first isolate the estimate used whenever an original character is exposed at the closing functional. The input can be an unreduced product: it is its actual analytic representative that enters the hypothesis.

Lemma 6.1 (Exact Fourier and sampling formulas)**.** *For any polynomial F, write its finite symmetric Laurent expansion*

$$F(w + w^{-1}) = \sum_{r\in\mathbb{Z}} a_r w^r, \qquad a_{-r} = a_r.$$

For $0 \le h \le d$,

$$\phi(L_{N-h}F) = (-1)^h N^2 \sum_{m\in\mathbb{Z}} (-1)^m (m+1) a_{h+mN}. \tag{6.1}$$

In particular the winding $m = -1$ has exactly zero coefficient. With $\theta_l = (2l+1)\pi/N$ and $w_l = e^{\mathrm{i}\theta_l}$, the same functional is

$$(-1)^h \sum_{l=0}^{N-1} w_l^{-h} \big[(N-h)F(c_l) + (w_l - w_l^{-1})F'(c_l)\big]. \tag{6.2}$$

Proof. The coefficient of z^d after reduction is the coefficient of z^{-1} in $L_{N-h}(z)F(z)/P_N(z)$ at infinity. Make the change $z = w + w^{-1}$. Then

$$P_N(w + w^{-1}) = w^{-N}(w^N + 1)^2, \qquad U_{N-h-1}\big((w + w^{-1})/2\big) = \frac{w^{N-h} - w^{-(N-h)}}{w - w^{-1}}.$$

After including $\mathrm{d}z = (1 - w^{-2})\,\mathrm{d}w$ and the signs from (2.2), the answer is $(-1)^h N^2$ times the coefficient of w^{-1} in

$$F(w + w^{-1}) \frac{w^{2N-h-1} - w^{h-1}}{(w^N + 1)^2}.$$

Expanding the denominator at infinity gives

$$\sum_{m\ge 0} (-1)^m (m+1)\{a_{h+mN} - a_{2N-h+mN}\}.$$

Use symmetry to rewrite the second half with negative windings. Its first winding is -2, leaving precisely the zero coefficient at -1 in (6.1).

Discrete orthogonality gives

$$\frac{1}{N}\sum_{l} w_l^{-h}\sum_{r} a_r w_l^r = \sum_{m}(-1)^m a_{h+mN}.$$

Applying $w\partial_w$ supplies the additional multiplier $h+mN$. The combination $(N-h)+w\partial_w$ is $N(m+1)$ on that frequency, proving (6.2). It counts every double node twice and the odd simple node once; the derivative multiplier is zero at the latter. Thus the formula retains exactly the full quotient data. □

If $\deg F < N$, only the term $m=0$ can survive in (6.1). For a product of canonical polynomials the other windings can occur, and must be kept.

Theorem 6.2 (Color loss from a spectral strip)**.** *Let E_N be actual polynomials, not necessarily canonical. Suppose there are constants $c>0, C>0, y_0>0$, a real number F, and positive A_N with $\log A_N = o(N)$, such that*

$$|E_N(2\cos(\theta+\mathrm{i}y))| \le A_N \exp N\{F - c(\theta-\pi)^2 + Cy^2\} \tag{6.3}$$

for $0\le\theta\le 2\pi$, $|y|\le y_0$. The constants and the error are uniform over any further specified parameters of E_N. Then, uniformly in j and $\epsilon\in\{0,1,-1\}$,

$$|\phi(R^\epsilon L_j E_N)| \le e^{N(F+o(1))}.$$

For each fixed $\delta>0$, the subfamily $N-j\ge\delta N$ satisfies

$$|\phi(R^\epsilon L_j E_N)| \le e^{N(F-\gamma_\delta)} \tag{6.4}$$

for some $\gamma_\delta>0$ and every sufficiently large N.

Proof. The proof has to retain the crossing derivative as well as its values. For $\epsilon\in\{0,1,-1\}$, define the entire function

$$\rho_{\epsilon,N}(z) = \kappa^\epsilon \exp\left[\mathrm{i}\epsilon\left(\frac{\pi}{2} - \frac{\pi}{4N} - \frac{N\pi}{4}\right)\right]\exp\left[\frac{\mathrm{i}\epsilon N}{4\pi}(z-\pi)^2\right]. \tag{6.5}$$

At every θ_l it equals $R^\epsilon(c_l)$. Its logarithmic derivative is $\mathrm{i}\epsilon N(\theta_l-\pi)/(2\pi)$, which is also the derivative of $R^\epsilon(2\cos z)$ there by (2.10). At an odd simple node only the value is needed. Hence, if $F_N(z)=\rho_{\epsilon,N}(z)E_N(2\cos z)$ and $h=N-j$, the exact functional is

$$Z_{N,h} = (-1)^h \sum_{l=0}^{N-1} e^{-\mathrm{i}h\theta_l}\{(N-h)F_N(\theta_l) - \mathrm{i}F_N'(\theta_l)\}. \tag{6.6}$$

The real-axis chirp has modulus one. Cauchy's inequality on radius $1/N$ disks bounds the derivative by a polynomial cost times $A_N e^{NF}$, proving the first conclusion. At endpoint disks use the periodicity of $E_N(2\cos z)$; its continuation still has the bound required near the endpoints.

For the strict loss, put $u=\theta-\pi$. Completing the square in the product of (6.3) and (6.5) gives, on a smaller fixed strip,

$$|F_N(\theta+\mathrm{i}y)| \le A_N e^{N(F-c_1u^2+C_1y^2)}, \qquad c_1>0. \tag{6.7}$$

For instance absorb $|uy|/(2\pi)$ into half the negative cu^2 term and a constant times y^2. On small fixed disks about $0, 2\pi$, the first several derivatives consequently have the common bound

$$CN^A A_N e^{N(F-\gamma_0)}, \qquad \gamma_0 > 0. \tag{6.8}$$

Choose those disks first and use Cauchy on smaller disks. Their distance from π is bounded below, so shrinking the strip leaves a fixed loss.

The function F_N is symmetric under $z \mapsto 2\pi - z$, but its derivatives do not all match at the two ends of the interval. Specifically,

$$F_N'(2\pi) - F_N'(0) = \mathrm{i}\epsilon N F_N(0).$$

Set

$$\psi(z) = z(2\pi - z), \quad b_N = -\frac{\mathrm{i}\epsilon N F_N(0)}{4\pi}, \quad G_N(z) = F_N(z) - b_N\psi(z). \tag{6.9}$$

The endpoint values of G_N, G_N', G_N'' match: the first derivatives match by the displayed choice of b_N, and the even derivatives match by reflection. The periodic extension is therefore C^2. Four integrations by parts, retaining the third-derivative jump, show that its Fourier coefficients

$$g_{N,k} = \frac{1}{2\pi}\int_0^{2\pi} G_N(\theta) e^{-\mathrm{i}k\theta}\,\mathrm{d}\theta$$

are $O_N((1+|k|)^{-4})$. In particular both its Fourier series and the differentiated series converge absolutely. Substitution in (6.6) is legitimate and gives

$$Z_{N,h}(G_N) = (-1)^h N^2 \sum_{m\in\mathbb{Z}} (-1)^m (m+1) g_{N,h+mN}. \tag{6.10}$$

The subtracted polynomial is not omitted. Its original nodal functional costs at most a polynomial times the bound (6.8); it will be restored at the end.

For $k > 0$, move the coefficient contour to imaginary part $-\eta$, with $0 < \eta$ fixed and small. For $k < 0$, move it to $+\eta$. There are two vertical joining sides. Their difference, for $k > 0$, contains

$$D_N(y) = G_N(2\pi - \mathrm{i}y) - G_N(-\mathrm{i}y), \qquad 0 \le y \le \eta.$$

By the endpoint matching, $D_N(0) = D_N'(0) = D_N''(0) = 0$. Three integrations by parts in $\int_0^\eta e^{-ky} D_N(y)\,\mathrm{d}y$ bound the joining contribution by

$$CN^A A_N e^{N(F-\gamma_0)} \left[e^{-k\eta}(k^{-1} + k^{-2} + k^{-3}) + k^{-4} \right].$$

The terms at η are the joining-side boundary terms; none is set to zero. The remainder uses the bound for D_N''' in (6.8). The horizontal segment of F_N is bounded by (6.7); the subtracted polynomial $b_N\psi$ has the smaller endpoint rate (6.8), multiplied by the same factor $e^{-|k|\eta}$. The same argument in the other half-plane proves, for $k \neq 0$,

$$|g_{N,k}| \le CN^A A_N \left\{ e^{N(F+C_1\eta^2)-|k|\eta} + \frac{e^{N(F-\gamma_0)}}{(1+|k|)^4} \right\}. \tag{6.11}$$

If $\delta N \le h \le N-1$, every frequency in (6.10) with nonzero weight has absolute value at least δN. The potentially small frequency $h - N$ has the zero weight $m + 1 = 0$. The exponential terms in (6.11), multiplied by $|m+1|$, sum to $O(e^{-\delta N\eta})$. The inverse fourth powers are likewise summable with that multiplier. Choose η so small that $C_1\eta^2 < \delta\eta/2$, then restore the correction in (6.9). Reducing the smaller positive loss absorbs $\log A_N = o(N)$ and all powers of N, proving (6.4). □

7 Multiplying by a character before turning

For an input class E, define

$$D_j^\epsilon[E] = \mathcal{T}(R^\epsilon L_j E). \tag{7.1}$$

The character stays at its specified position before the turn. The input may be a product of several subtree polynomials; it need not have a positive coefficient expansion.

Lemma 7.1 (Coefficient duality)**.** *For $0 \le r \le d$,*

$$[e_r]D_j^\epsilon[E] = \phi(e_r R^\epsilon L_j E), \qquad |[e_r]D_j^\epsilon[E]| \le C N^A \|E\|_{\mathcal{J}} w_{N,r}. \tag{7.2}$$

Proof. The pairing in (2.2) extracts the coefficient $d-r$ before the turn, hence coefficient r after the turn. Use (2.19), submultiplicativity and (2.20). There is no sum over the coefficients of the input representative. □

Set $G_0 = g(1/4)$. Differentiation gives

$$g'(u) = 2\log\cot(\pi u), \qquad g''(u) = -\frac{4\pi}{\sin(2\pi u)} < 0, \qquad 0 < u < 1/2. \tag{7.3}$$

Thus $G_0 > 0$ is the unique maximum of g, while $g(0) = g(1/2) = 0$.

Lemma 7.2 (Angular bound for a low Newton polynomial)**.** *Let I be a compact subinterval of $(0, 1/2)$. There are $c_I, C_I, y_I > 0$ and $e_N \to 0$ such that, whenever $r/N \in I$,*

$$|e_r(2\cos(\theta + \mathrm{i}y))| \le w_{N,r} \exp N\{-c_I(\theta-\pi)^2 + C_I y^2 + e_N\} \tag{7.4}$$

for $0 \le \theta \le 2\pi$, $|y| \le y_I$. On the full low range, the upper logarithmic rate is

$$b(u,\theta,y) = \int_0^u \log|2\cos(2\pi v) - 2\cos(\theta + \mathrm{i}y)|\,\mathrm{d}v - 2J(u). \tag{7.5}$$

At $y = 0$, $g(u) + b(u,\theta,0)$ has the unique maximum $2G_0$ at $(u,\theta) = (1/4, \pi)$.

Proof. The numerator product in (2.1) is a midpoint Riemann sum for the integral in (7.5). Factor each cosine difference into its two sine factors. For an upper bound, replace each of their logarithms separately by its maximum with $-K$. The resulting continuous functions have uniform Riemann sums on the compact parameter set. The negative tails of the two circular logarithms have integrals bounded uniformly by $Ce^{-K}(1+K)$: near a zero this follows by integrating $|\log|t||$ on a short interval, and moving the zero off the real circle only decreases its negative tail. This also covers coalescing zeros. Let $K \to \infty$ after taking the upper limit in N. The factorial estimate in equation (2.16) handles the denominator, including both endpoints. This proves a uniform upper error $o(1)$ in the rate.

Here is the strict real comparison. Write $\ell(t) = \log|2\sin(t/2)|$, a periodic function of period 2π. For $0 < u < 1/2$, the numerator integral is

$$\frac{1}{2\pi}\int_{\theta-2\pi u}^{\theta+2\pi u} \ell(t)\,\mathrm{d}t.$$

As the center θ moves from 0 to π, this integral strictly increases. Its derivative is the difference of its endpoint values divided by 2π; that difference is positive because

$$\sin^2\frac{\theta+\alpha}{2} - \sin^2\frac{\theta-\alpha}{2} = \sin\theta\sin\alpha > 0, \qquad \alpha = 2\pi u \in (0,\pi).$$

The isolated logarithmic singularities are integrable, so the argument extends across them by continuity. Reflection treats $\pi < \theta < 2\pi$. It follows that $b(u,\theta,0) \le b(u,\pi,0) = g(u)$, strictly away from π when $0 < u < 1/2$. At $u = 0, 1/2$ the rate is zero. Equation (7.3) proves the stated unique joint maximum.

Near $\theta = \pi$, put $w = \theta - \pi + \mathrm{i}y$. The numerator has no zeros for $u \in I$ in a small fixed neighborhood, and its holomorphic action there is

$$B(u,w) = -J(1/2 - u + w/(2\pi)) - J(1/2 - u - w/(2\pi)) - 2J(u). \tag{7.6}$$

The analytic branch of J is the real one on this compact interior interval. It satisfies

$$B(u,0) = g(u), \quad B_w(u,0) = 0, \quad B_{ww}(u,0) = -\frac{\tan \pi u}{2\pi} < 0.$$

Uniform Taylor bounds give a negative angular quadratic and a positive quadratic height cost there. The remaining real compact set has a strict gap by the preceding comparison. Continuity of the logarithmic integrals preserves half that gap at sufficiently small fixed heights. Reducing the angular constant gives (7.4) on the whole strip. Finally use (2.21) to replace $e^{Ng(r/N)}$ by $w_{N,r}$. □

Proposition 7.3 (Norm and angular localization after a character)**.** *Suppose* $\|E_N\|_{\mathcal{J}} \le e^{N(c+o(1))}$*, uniformly in the specified input family. Then*

$$\|D_j^{\epsilon}[E_N]\|_{\mathcal{J}} \le e^{N(c+2G_0+o(1))}. \tag{7.7}$$

Its canonical representative has constants $a, C, y_0 > 0$ *such that*

$$|D_j^{\epsilon}[E_N](2\cos(\theta + \mathrm{i}y))| \le e^{N(c+2G_0 - a(\theta-\pi)^2 + Cy^2 + o(1))} \tag{7.8}$$

on $0 \le \theta \le 2\pi$, $|y| \le y_0$*. The constants are independent of the physical color* j *and retained sign. A fixed loss in the norm exponent of* E_N *is preserved in the coefficient estimate* (7.2) *at every output index.*

Proof. For low indices, reconstruct the polynomial using (7.2) and (2.20). The norm is bounded by a polynomial times $\|E_N\|_{\mathcal{J}} \sum_{r \le M} w_{N,r}^2$, whose rate is at most $c + 2G_0$. The high part has rate at most c, proving (7.7).

For the low angular estimate, the rate of the r-th term is $c + g(r/N) + b(r/N, \theta, y)$. Lemma 7.2 gives a strict index and angular quadratic at its unique maximum and a strict gap on the compact complement. The same argument with a small fixed height gives the required bound for this entire low sum. The high part is itself canonical, has norm at most $e^{N(c+o(1))}$, and by (2.22) has strip rate at most $c + |y|$. Choose y_0 and then the angular constant small enough that $y_0 + a\pi^2 < 2G_0$. This part then lies below (7.8) with a fixed loss. All sums have at most N terms. The final assertion follows directly from (7.2). □

Proposition 7.4 (A deficit in the new character)**.** *Suppose an actual representative of* E_N *satisfies*

$$|E_N(2\cos(\theta + \mathrm{i}y))| \le e^{N(c - c_E(\theta-\pi)^2 + C_E y^2 + o(1))}, \quad |y| \le y_E, \tag{7.9}$$

with $c_E \ge 0$*. The empty input* $E_N = 1, c = 0$ *is allowed. For every compact* $I \subset (0, 1/2)$ *and every* $\delta > 0$*, there is* $\eta > 0$ *such that*

$$N - j \ge \delta N, \quad r/N \in I \quad \Longrightarrow \quad |[e_r] D_j^{\epsilon}[E_N]| \le e^{N(c + g(r/N) - \eta)} \tag{7.10}$$

for all sufficiently large N*. If also* $\|E_N\|_{\mathcal{J}} \le e^{N(c+o(1))}$*, then this subfamily has* $\|D_j^{\epsilon}[E_N]\|_{\mathcal{J}} \le e^{N(c+2G_0-\eta')}$ *for some* $\eta' > 0$*.*

Proof. By Lemma 7.2, the actual product $e_r E_N$ satisfies the positive-angular hypothesis of Theorem 6.2 uniformly for $r/N \in I$, at rate $c + g(r/N)$. The first identity in (7.2) and that theorem prove (7.10). No continuation of the discrete color j is used.

For the norm assertion take I to be a small interval about $1/4$. Its contribution loses η. The other low indices have a fixed loss from $2g(u) < 2G_0$ outside I, and the high part has the fixed gap $2G_0$. Apply (7.2) to those remaining terms and sum. Reducing the minimum loss proves the claim. □

For a later component, the relevant input index can be strictly above $1/4$. This is why (7.10) is stated on every compact regular interval, rather than only near the maximum of g.

8 A positive comparison for a region without a character

Consider a connected region of product junctions with no distinguished original character at a junction. Each internal edge carries a turn. Choose one outgoing edge. Every junction then has at least two incoming branches. Its boundary inputs are $n \geq 2$ polynomials $F_{i,N}$ of type L. The resulting expression, before its outgoing turn, is denoted by $A(F_1, \dots, F_n)$, and its outgoing operator by $Q_N = \mathcal{T}A(F_1, \dots, F_n)$.

The finite expression uses only products and turns; every input occurs once. It can be constructed from two-input products by turns, products of completed expressions and single-input attachments. Its type refers to its pure positive states, not to the range of its signed Newton coefficients.

Definition 8.1 (The comparison arrays)**.** *Let $c_i \in \mathbb{R}$ be fixed constants. The input arrays satisfy the comparison hypothesis if a common $e_N \to 0$ gives*

$$|[e_r]F_{i,N}| \leq e^{N(c_i+e_N)} w_{N,r}, \qquad 0 \leq r \leq d, \quad 1 \leq i \leq n. \tag{8.1}$$

Replace every input coefficient by its absolute value, and apply the same products and turns; call the resulting operator A_N^{abs}. It is a positive coefficient comparison, not the pointwise absolute value of the polynomial A_N.

Introduce the auxiliary polynomial

$$W_N = \sum_{r=0}^{d} w_{N,r} e_r. \tag{8.2}$$

It serves only in a norm inequality. It does not add a crossing circle to the link. A *band sector* chooses, for each input and for this comparison polynomial, either the low indices $r \leq M$ or the high indices $r > M$. These bands are not the physical core and alias.

Let A^0 be the same expression with every input replaced by the pure complete leaf $h_N = \mathcal{T}L_N$. Write Φ_A for the complete action of the closed reference expression $\phi(h_N A^0)$, and set

$$\Omega = \sum_i c_i + \Phi_A. \tag{8.3}$$

The reference has $n + 1 \geq 3$ effective leaves. It has no flagged junction and is strict; these facts and its stationary boundary indices are supplied by [3].

Theorem 8.2 (Positive comparison with shared indices)**.** *For the fixed component and inputs (8.1),*

$$\|Q_N\|_{\mathcal{J}} \leq C N^A \phi(W_N A_N^{\mathrm{abs}}) \leq e^{N(\Omega + o(1))}. \tag{8.4}$$

Every band sector with a high input or a high comparison index has upper bound $e^{N(\Omega-\gamma)}$, for some $\gamma > 0$ and all sufficiently large N. The constants are uniform over the specified input families.

The all-low comparison is concentrated at the unique strict maximum of the original reference action. Its effective input indices and comparison index are in $(1/4, 1/2)$. Outside any fixed neighborhood of that maximum it has a fixed exponential loss.

Proof. Positivity of (2.4) and exact coefficient reversal give $|[e_r]A_N| \le [e_r]A_N^{\mathrm{abs}}$. By the full pairing, $[e_r]Q_N = \phi(e_r A_N)$. Hence

$$\sum_r w_{N,r}|[e_r]Q_N| \le \phi(W_N A_N^{\mathrm{abs}}). \tag{8.5}$$

This proves the first inequality in (8.4) by (2.20). It holds separately in each specified band sector and retains the first derivatives of all high basis elements.

It remains to bound the positive scalar. For low input and comparison indices its upper leaf profile is g; for high indices it is zero. The common additive constants are $\sum_i c_i$. Set

$$\widetilde{g}(x) = g(\max(x, 1/4)), \quad 0 \le x \le 1/2, \qquad k(x) = -2J(x/2), \quad 0 \le x \le 1.$$

On these domains $g \le \widetilde{g}$ and $0 \le k$. Thus the scalar supporting-plane estimates of Lemmas 4.1 and 5.1 apply to the profiles of (8.1). We give the propagation argument to specify why their applicability does not depend on the origin of the arrays.

First suppose $n + 1 \ge 4$. The unflagged closed junction tree can be rooted between two completed expressions. If there is an internal edge, each side has at least two leaves, because every internal vertex has valence at least three. If there is only one junction, split its at least four leaves into two products with at least two factors each. Products, associativity and (2.3) realize this change of root as an exact identity of full colored tensors. Every external band stays attached to its original array. No tangle isotopy is asserted.

At a two-leaf seed with selected output b, write its pure profile as $f(b)$, with slope $\lambda = f'(b)$. The effective boundary parameters have $1/2 < X < 1$. The two-leaf scalar inequalities give

$$\text{nonempty high profile at output } z \le f(b) + \lambda(z - b) - 1/100. \tag{8.6}$$

The all-low case has the same tangent without a loss. At a turn the output index changes from r to $d - r$, so the tangent slope changes sign. At a product of two completed children, add their affine bounds to the supporting plane of the concave function m. Stationarity cancels both child slopes. The outgoing slope is the derivative of the parent profile, and inherited losses add.

For a single-input attachment, use the child's entire affine bound, not an elementary-leaf domain for that child. If the new input is low, the tangent of $\widetilde{g}$ preserves the child's loss. If it is high, the single-attachment inequality supplies a further loss $1/2500$. Its hypotheses are the effective stationary inequalities $1/2 < X < 1$ and $0 < Y, Y_o < 1$ from the pure completed construction. These hypotheses concern the fixed reference profiles. They impose no additional condition on the coefficient arrays being bounded.

At closure the two reference outputs sum to one and their slopes agree. The exact finite indices sum to $d/N = 1 - 1/N$, so the remaining affine term is $-\lambda/N$. After multiplication by N, it is a bounded constant. Every nonempty high assignment has inherited at least one of the positive losses above. This proves the required strict rate bound for $n + 1 \ge 4$.

For $n = 2$, the closed reference has three effective leaves. There is a shorter direct argument. Factorial reflection $S_r S_{d-r} = N$ in (2.4) gives the symmetric tensor

$$\phi(e_i e_j e_k) = \frac{S_{i+j} S_{i+k} S_{j+k}}{N S_i S_j S_k S_{i+j+k-d}} \tag{8.7}$$

when $i+j, i+k, j+k \le d$ and $i+j+k \ge d$, and zero otherwise. In particular two high indices force the coefficient to vanish: their sum is at least $2M+2 > d$, at either parity. For exactly one high index, group it with a low index, leaving the other low index for closure. The pure selected point is

$$a = \frac{\arctan 2}{\pi}, \quad b = 1-a, \quad X = 1/2, \quad \lambda = g'(a) = -\log 4.$$

The endpoint case of the two-leaf inequality gives $f(b) + \lambda(z-b) - 1/100$. The remaining low profile has tangent $g(a)+\lambda(u-a)$. Since $z+u = d/N$, their sum is at most $\Phi_A - 1/100 + \log 4/N$. This proves the high-sector claim also in the three-leaf reference. No estimate for three physical aliases is needed here: these are coefficient bands, and the two-high vanishing is an exact polynomial identity.

For the all-low sector, the uniform closed upper action is

$$\sum_i c_i + g(u) + \sum_i g(a_i) + \sum_v m(x_v, y_v, z_v), \tag{8.8}$$

with the reference's original shared indices. Its unique strict maximum is Ω. The stationary input and comparison indices lie in $(1/4, 1/2)$, including the value a just displayed in the three-leaf case. These are the strict optimizer and effective boundary-range results of [3]. Compactness gives a strict loss outside each fixed neighborhood of that maximum.

There are only polynomially many finite states, because the component is fixed. The factorial rates are uniform on the whole closed support. Any limit of finite states satisfies the limiting turn and closing constraints, so continuity of the actions transfers the compact gaps to the displaced finite grids. Powers of N, the common error in (8.1) and the fixed number of sectors are absorbed by reducing the loss. This proves all the assertions. □

Corollary 8.3 (Transfer of an input coefficient loss)**.** *Let a_i^* be the selected index of input i in (8.8). Suppose that, on a fixed interval I_i containing a_i^* in its interior, a specified subfamily improves (8.1) to*

$$|[e_r]F_{i,N}| \le e^{N(c_i + g(r/N) - \eta + o(1))}, \qquad r/N \in I_i, \quad \eta > 0. \tag{8.9}$$

Then $\|Q_N\|_{\mathcal{J}} \le e^{N(\Omega - \eta')}$ for some $\eta' > 0$ on that subfamily, for every sufficiently large N.

Proof. The high sectors already have a fixed gap. In the all-low sector, choose a neighborhood of the strict maximum whose i-th coordinate lies in I_i. Inside it the improved coefficient removes η from every state rate; outside it compactness gives a gap. Use (8.5) and reduce the minimum of these losses. The arrays can be complex or identically zero; no lower bound for them is used. □

Proposition 8.4 (Angular localization at the reference rate)**.** *The canonical outgoing polynomial in Theorem 8.2 satisfies*

$$|Q_N(2\cos(\theta + \mathrm{i}y))| \le e^{N(\Omega - c(\theta-\pi)^2 + Cy^2 + o(1))} \tag{8.10}$$

on a fixed strip, with $c, C > 0$, uniformly in its physical inputs.

Proof. Choose a compact interval $I \subset (0, 1/2)$ around the selected comparison index. Separate the actual state expansion of Q_N into an all-low part with output index $r/N \in I$, and its complement. The weighted coefficient mass of the complementary part has a strict gap by Theorem 8.2. It is a canonical polynomial, so (2.22) bounds its strip rate by $\Omega - \gamma + |y|$.

For the first part apply (7.4) to each output e_r. Its factor $w_{N,r}$ is already present in (8.5); no further exponential cost is added. This gives (8.10) for that part. Finally choose $y_0 < \gamma/4$ and $c\pi^2 < \gamma/4$, reducing the angular constant if needed. The complementary part then fits under the same bound. The two contributions together prove the assertion. □

9 Induction over the original composition

We now explain why the two estimates just proved exhaust a strict source, and why their exponents add to its own complete action. The reduction is an equality of finite contractions. The original ordered diagram is still needed for any later geometric interpretation.

9.1 The reduced tree and its constants

Associate consecutive products and cancel pairs of adjacent turns using $\mathcal{T}^2 = 1$ and (2.3). The result is a tree of product junctions, each of valence at least three. Every internal edge carries a turn. An external edge carries either $h_i = \mathcal{T}(R^{\epsilon_i} L_{j_i})$, or its turn $p_i = R^{\epsilon_i} L_{j_i}$. Call the latter edge *flagged*. The reduced-junction theorem in [3] says that strictness is equivalent to at most one flagged external edge at each junction. All labels and full polynomial classes remain in this tree.

A junction with a flagged leaf sends toward its parent an operator

$$F_i = D_{j_i}^{\epsilon_i}[E_i], \qquad E_i = \prod_b Q_b. \tag{9.1}$$

An ordinary external leaf is the same operation with $E_i = 1$. These are the type-L modules used at the boundaries of Section 8. Every maximal connected set of unflagged junctions gives the other operation,

$$Q = \mathcal{T}A(F_1, \dots, F_n), \qquad n \geq 2. \tag{9.2}$$

Its boundary modules are all type L; adjoining unflagged junctions were already included in the same maximal component. Both operations depend only on smaller subtrees.

Let $f_Q(b)$ be the optimized pure action with output b held fixed, and define its intrinsic rate

$$\Omega_Q = \max_{0 \leq b \leq 1/2} \{g(b) + f_Q(b)\}.$$

The harmonic-profile and probe theorems of [3] give

$$f_{F_i}(b) = g(b) + c_i, \qquad c_i = \sum_b \Omega_{Q_b}, \qquad \Omega_{F_i} = c_i + 2G_0, \tag{9.3}$$

$$\Omega_{\mathcal{T}A(F_1,\dots,F_n)} = \sum_i c_i + \Phi_A. \tag{9.4}$$

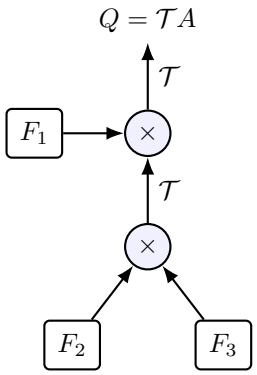
The sum in (9.3) is zero for an empty input. These equalities include the additive constants, not just derivatives of profiles. The quantity in (9.4) is precisely (8.3). Thus the positive comparison has not introduced an extra exponential budget.

9.2 Properties propagated from the leaves

We prove three properties together. Every subtree output Q has the norm bound $e^{N(\Omega_Q + o(1))}$, a canonical angular strip bound at that same rate with a strictly negative angular quadratic, and a fixed norm loss whenever any one of its original colors has a macroscopic deficit. In addition, a type-L module F_i has the coefficient envelope (8.1), and each original color inside that module gives a strict coefficient loss on every fixed compact interval in $(0, 1/2)$.

For an elementary leaf, these assertions are Lemma 7.1, Proposition 7.3 and Proposition 7.4, with $E = 1$. In particular its coefficient reference constant is zero and its norm rate is $2G_0$.

Suppose next that F_i is given by (9.1). Multiply the canonical representatives of the Q_b without first reducing that product. The norm bounds multiply by submultiplicativity, their



$$W_N = \sum_r w_{N,r} e_r$$

$$\|Q_N\|_{\mathcal{J}} \le C N^A \phi(W_N A_N^{\mathrm{abs}})$$

A_N^{abs} uses the same products and turns, with the absolute values of the boundary coefficients.

Figure 2: An unflagged region with three boundary modules. Here $A = F_1\mathcal{T}(F_2F_3)$ and $Q = \mathcal{T}A$. The internal turn is part of the original expression. The positive scalar on the right bounds its outgoing norm with the same shared indices. The polynomial W_N belongs to this inequality only.

analytic strip bounds multiply as actual polynomials, and their rates sum to c_i in (9.3). A deficit in a descendant gives a strict norm loss in this product. Lemma 7.1 then preserves that loss in every output coefficient. A deficit in the newly inserted character gives the regular-index coefficient loss by Proposition 7.4. That proposition also supplies the norm loss for the new color. Proposition 7.3 gives the canonical strip estimate at rate $c_i + 2G_0 = \Omega_{F_i}$. This proves every claimed property for the new type-L module.

For (9.2), the input coefficient envelopes give Theorem 8.2, and Proposition 8.4 regenerates the angular estimate at Ω_Q. A deficit in any original descendant lies inside one boundary module F_i. Its coefficient loss holds on a compact interval about the actual selected index a_i^*, even when that index is above $1/4$. Corollary 8.3 gives the required norm loss in Q. Equation (9.4) identifies its rate with the intrinsic rate of this very subtree.

This is an induction on a finite tree. At each step the constants may depend on the subtree, but not on its physical color tuple. Taking the minimum of finitely many positive losses leaves a positive loss. No lower estimate or nonzero value of an arbitrary subtree is assumed.

9.3 The closing character, including an unflagged root

The arbitrary-leaf rooting identity of [3] allows one original leaf to be exposed without changing the contraction. If its edge parity is $\sigma \in \{0, 1\}$, and A is the product of the other branches at its junction, then

$$Z_N = \phi((\mathcal{T}^\sigma h_0)A) = \phi(R^{\epsilon_0} L_{j_0} \mathcal{T}^{\sigma+1} A). \tag{9.5}$$

There are at least two other branches because $H \ge 3$ and junctions have valence at least three.

If $\sigma = 1$, the exposed leaf is the sole possible flag there. The final input in (9.5) is the unturned product of the remaining branches. Its norm rate is their sum of intrinsic rates, which the root-budget identity of [3] identifies with F_0. Their strip estimates multiply to give a positive angular quadratic at that rate.

If $\sigma = 0$ and another leaf at this junction is flagged, the final input is the type-L module supplied by that other character. If no leaf there is flagged, it is the outgoing turn of an unflagged component. Both cases have already been treated by the induction, including their angular estimates. The same root-budget identity says $\Omega_{\mathcal{T}A} = F_0$. One can also see this

equality directly from the positive closing constraint:

$$F_0 = \max_a \{g(a) + f_A(1-a)\} = \Omega_{\mathcal{T}A}.$$

Now apply (2.19) to bound the full contraction. A deficit in a descendant gives a strict norm loss by the induction; the closing character has only polynomial norm cost. A deficit in the exposed color j_0 is handled by Theorem 6.2 on the inherited strip. This proves (1.2) and (1.3). It also treats a tree with no flagged junction anywhere. In particular the three-leaf closure uses exactly the $n = 2$ comparison in Theorem 8.2; it requires no additional source theorem.

9.4 From a physical color loss to the core comparison

For a physical high color $j > N/2$, the exact circle formula gives

$$h_j^\epsilon = C_j^\epsilon + (-1)^N h_{N-j}^\epsilon, \qquad h_0^\epsilon = 0. \tag{9.6}$$

Here C_j^ϵ is the selected core; set it equal to zero for $j \le N/2$. Formula (9.6) follows from the pure complementary-color identity of [1, Section 4] by the linear operation $F \mapsto \mathcal{T}(R^\epsilon \mathcal{T} F)$. It therefore retains the crossing and all repeated-root derivatives.

If some original color is at most $N/2$, the all-core contraction is zero, and (1.3) with $\delta = 1/2$ proves (1.4). Otherwise replace each core in its multilinear contraction by $h_j^\epsilon - (-1)^N h_{N-j}^\epsilon$. Subtracting from the full source leaves finitely many terms with at least one color $N - j \le N/2$. Terms containing color zero vanish. Every other term is an original physical source with a macroscopic color deficit, so (1.3) applies. The number of terms is at most $2^H - 1$, independent of N. Reducing the gap absorbs that finite sum and proves (1.4), completing Theorem 1.1.

9.5 The strict two-leaf closure

Corollary 9.1 (Two leaves with their original signs)**.** *Let $H = 2$ and suppose the closure is strict. After the exact turn and pairing identities it is*

$$Z_N = \phi(h_{j_0}^{\epsilon_0} \mathcal{T} h_{j_1}^{\epsilon_1}) = \phi(p_0 \mathcal{T} p_1), \qquad p_i = R^{\epsilon_i} L_{j_i}, \quad \epsilon_i \in \{0, 1, -1\}.$$

Its pure complete rate is $F_0 = 2G_0$. The whole-source bound (1.2), every macroscopic physical-color gap (1.3), and the full-minus-core gap (1.4) hold with this value, at all sufficiently large integer levels. No unflagged-component comparison is needed.

Proof. Strictness leaves one turn between the two elementary leaves, modulo $\mathcal{T}^2 = 1$ and the adjoint pairing identity. The displayed equality then follows from $h_j^\epsilon = \mathcal{T} p_j$. For the pure complete circle, the exact coefficients in (2.7) give

$$h_N = \sum_{r=0}^{M} a_{N,r} e_r, \qquad a_{N,r} = \frac{N^2 S_{2r}}{S_r^4}, \qquad \phi(h_N \mathcal{T} h_N) = \sum_{r=0}^{M} a_{N,r}^2.$$

Thus the normalization is the original coefficient-square sum, without an extra trace factor. Uniform factorial rates and one index nearest $N/4$ give its all-level exponential rate $2G_0$.

The final input to the character p_0 is exactly $\mathcal{T} p_1 = D_{j_1}^{\epsilon_1}[1]$. Proposition 7.3 gives its norm and angular strip at rate $2G_0$. A deficit in j_1 gives a norm gap by Proposition 7.4; the other character then has only the polynomial cost (2.19). A deficit in j_0 gives a gap by Theorem 6.2 on that same strip. The norm bound also gives the full upper rate. Finally the exact physical alias expansion from the preceding subsection converts these color gaps into the full-minus-core gap. It preserves both retained crossing signs and all double-root derivatives. □

10 Exact rational verification of the two comparisons

The two finite supporting-plane arguments use fixed rational points, not numerically located maximizers as hypotheses. This section specifies the data and proves how every transcendental quantity in the comparisons is enclosed. Together with concavity and the exact covers in Sections 4 and 5, these are finite proofs of the two inequalities.

The ancillary files in the repository's `experiments/` directory are

```
conway_two_leaf_block_alias_certificate.json
conway_single_leaf_attachment_certificate.json
verify_conway_two_leaf_block_alias_certificate.py
verify_conway_single_leaf_attachment_certificate.py.
```

The first table records the sixteen slope intervals and three rational points in each. The second records a rooted subdivision tree; a node is either its four quarter-rectangles or one pure point and one alias point. Coordinates are integer multiples of the denominator declared by the table. The verifier checks strict feasibility and the core conditions before taking any derivative. For the first alias profile its core tangent coordinate is strictly between $1/4$ and $1/2$, so the derivative is the stated derivative of g.

10.1 Outward interval arithmetic

Let $S = 10^{40}$. An interval is stored as integers (l, u) representing $[l/S, u/S]$. A rational input is rounded down and up to this grid. Addition and subtraction are exact on the grid. Multiplication uses all four endpoint products with outward integer rounding; division uses all four endpoint quotients after checking that the denominator interval avoids zero. Thus every subsequent interval contains the exact expression, by induction on its arithmetic operations. Repeated multiplication gives nonnegative integer powers.

For the constant π use Machin's identity

$$\pi = 16\arctan(1/5) - 4\arctan(1/239).$$

Truncate each alternating series after 40 terms and enclose its remainder by the absolute next term. The resulting interval also verifies $\pi < 22/7$.

For $\sin(\pi t)$, reflect $t > 1/2$ to $1 - t$ and use the Taylor polynomial through degree 41. On $0 \le \pi t \le \pi/2$, its error is at most $(\pi t)^{43}/43!$. Rational interval operations applied to the enclosed π give an outward enclosure. Every sine used inside a logarithm is at a strict positive rational argument below one; positivity of its resulting enclosure is checked.

For a positive interval x, multiply by a power of two until its lower endpoint is in $[1, 2)$. Put $z = (x - 1)/(x + 1)$ and verify $0 \le z < 1/2$ throughout the interval. Then

$$\log x = 2\sum_{j=0}^{79} \frac{z^{2j+1}}{2j+1} + R, \qquad 0 \le R \le \frac{2z^{161}}{161(1 - z^2)}. \tag{10.1}$$

The upper bound follows by replacing every subsequent denominator by 161 and summing the geometric powers. The same formula at $z = 1/3$ encloses $\log 2$ and restores the extracted power. Monotonicity and outward rounding are used at every step; the small width of the interval is not presumed.

10.2 The log-sine integral and gradients

For $0 < t < 1/2$, integration of the absolutely convergent Euler sine product gives

$$J(t) = t\log(2\pi t) - t - \sum_{n=1}^{20} \frac{\zeta(2n)t^{2n+1}}{n(2n+1)} - R_t, \qquad 0 \le R_t \le \frac{2t^{43}}{21 \cdot 43(1 - t^2)}. \tag{10.2}$$

To justify the expansion directly, write $\log(\sin \pi u/(\pi u)) = \sum_{k\geq 1} \log(1 - u^2/k^2)$, expand each negative logarithm and integrate; the nonnegative double series permits termwise integration. For the tail use $\zeta(2n) < 2$ and $n(2n+1) \geq 21 \cdot 43$ for $n \geq 21$.

The first zeta value is $\zeta(2) = \pi^2/6$. For $2 \leq n \leq 20$ integral comparison gives

$$\sum_{k=1}^{64} k^{-2n} \leq \zeta(2n) \leq \sum_{k=1}^{64} k^{-2n} + \frac{64^{1-2n}}{2n-1}.$$

All these bounds are rational except for the already enclosed π. Use the exact values of J at $0, 1/2, 1$, and use $J(1-t) = -J(t)$ above $1/2$.

The gradients needed for the supporting planes contain only positive sines:

$$m_x = \log \frac{\sin \pi(x+y) \sin \pi(z-x)}{\sin \pi x \sin \pi(x+y-z)},$$
$$m_y = \log \frac{\sin \pi(x+y) \sin \pi(z-y)}{\sin \pi y \sin \pi(x+y-z)},$$
$$m_z = \log \frac{\sin \pi z \sin \pi(x+y-z)}{\sin \pi(z-x) \sin \pi(z-y)}.$$

Together with the displayed derivatives of g and k, equations (10.1) and (10.2) enclose every term of (4.5) and (5.5). The lower interval endpoints give precisely the rational margins (4.6) and (5.6). Evaluating a tangent at a boundary vertex requires no singular derivative at that vertex: all derivatives are taken at the strict rational tangent point.

The elementary bound (5.8) can also be checked without special-function evaluation. Its four-term lower sum is

$$\frac{2}{3} - \frac{(2/3)^3}{9} + \frac{(2/3)^5}{25} - \frac{(2/3)^7}{49} = \frac{1708762}{2679075}.$$

For $\log(3/2)$, the first three terms of its $z = 1/5$ series and the bound on its remaining positive tail give

$$\log(3/2) \leq 2\left(\frac{1}{5} + \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5}\right) + \frac{2}{5^7 \cdot 7(1 - 1/25)} = \frac{177391}{437500}.$$

Multiplying the first lower bound by $7/11$ and subtracting the second gives the fraction in (5.8).

10.3 Reproduction and the limits of the finite computation

From the repository root, with Python's standard library alone, run

```
python3 papers/conway-source-bounds/check.py \
    --output /tmp/conway-source-check
```

The wrapper replays the two certificates, verifies their coverage counts and margins, and checks the elementary rational comparison. It also calls `check_fourier.py`, which compares (6.1) with independent monic polynomial division at $N = 2, \ldots, 18$, including unreduced products and multiples of P_N. The same program checks the rational crossing phases and angular derivatives, the endpoint-jet repair, and the two-high support exclusion. Nonzero classes invisible to all spectral values and negative controls for a nonzero missing winding or deleted normals test the need to retain the full quotient.

The wrapper records source and data hashes before and after execution. It rejects optimized Python, runs isolated subprocesses and removes any old success summary before

validation. The output contains both detailed rational records, the bounded finite-algebra record and a summary.

The computation proves the finite inequalities once the enclosure rules and concavity arguments above have been established. It does not sample levels to prove the Ward identity, the long matrix-product bounds, or the tree induction. Those are all-level written arguments. The physical identification of the finite algebra remains the stated import from [1]. Neither the interval certificates nor the bounded Fourier checks replace that identification or prove an analytic estimate by samples.

The theorem can be used in a later filling argument in the following precise way. It supplies a strict bound on the sum of all alias sectors and on the entire physical-color complement of a fixed complete-color neighborhood. A filling analysis must still keep its exact finite functional, control the remaining neighborhood and prove a nonzero selected contribution with the appropriate normalization. These requirements are not conclusions of an absolute source bound.

References

[1] *Newton coefficients and positive sources for Conway tangles.* Companion paper, PDF.

[2] *Finite spectral evaluation and uniform filling bounds.* Companion paper, PDF.

[3] *Stationary profiles of strict Newton trees.* Companion paper, PDF.

[4] *Rational supporting-plane data for two-leaf blocks and single-leaf attachments.* Ancillary data and exact verifiers, repository experiments; the exact files and replay are specified in Section 10.